Optimal Market Making

Inventory control, exact quotes, signals, and estimating the fill curve

Yaohui Shu

Stony Brook University

October 2026

The question

A market maker earns the spread on each fill and carries the inventory the fills leave behind. **Where should it quote, given its inventory, its risk aversion, and how fills respond to price?**

- Too tight: many fills, large inventory swings.
- Too wide: few fills, little revenue.
- Skewed: lean against inventory to unload it.

The control problem has an exact solution when fills arrive at an exponential rate in the quote distance. Its inputs, the fill curve and the drift, must be estimated, and errors in them cost money.

In this deck

1. The Avellaneda-Stoikov problem and HJB equation.
2. The exact GLFT solution with inventory limits.
3. Quoting with a short-lived signal.
4. Estimating the fill curve, with censoring.
5. What a wrong fill curve costs.

The problem

Mid price $\mathrm{d}S_t = m\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, with drift m (ticks s^{-1}) and volatility σ (ticks $\mathrm{s}^{-1/2}$). The maker quotes a bid at $S - d^b$ and an ask at $S + d^a$ (distances in ticks). Each quote is filled at rate

$$\Lambda(d) = A\,e^{-kd},$$

with A (s^{-1}) the rate at the mid and k (tick^{-1}) the decay. Cash x, inventory y (shares, $|y| \le Q$), horizon T (s). Maximize

$$\mathbb{E}\big[-\exp\big(-\gamma(x_T + y_T S_T)\big)\big]$$

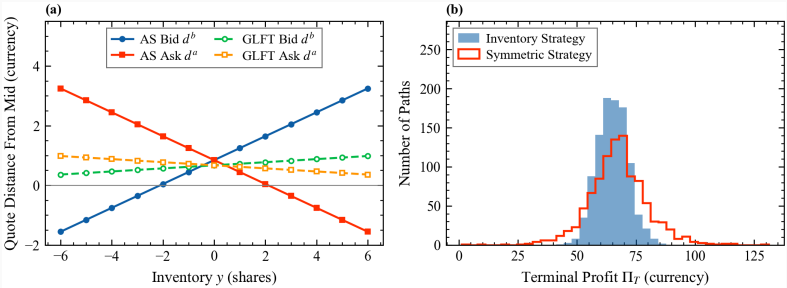
with risk aversion γ (tick^{-1} per share).

HJB equation for the value $u(t, x, y, S)$:

$$\begin{aligned} 0 = {} & \partial_t u + m\,\partial_S u + \tfrac{1}{2}\sigma^2 \partial_{SS} u \\ & + \sup_{d^b} A e^{-kd^b}\big[u(x - S + d^b, y + 1) - u\big] \\ & + \sup_{d^a} A e^{-kd^a}\big[u(x + S + d^a, y - 1) - u\big], \end{aligned}$$

with $u(T) = -e^{-\gamma(x+yS)}$. Avellaneda and Stoikov (2008) expand in y; Guéant et al. (2013) solve it exactly.

Avellaneda-Stoikov: reservation price and spread



To first order in inventory, quotes are symmetric around the reservation price r_t:

$$r_t = S_t - y\,\gamma\sigma^2(T-t), \qquad d^a + d^b = \gamma\sigma^2(T-t) + \frac{2}{\gamma}\ln\left(1+\frac{\gamma}{k}\right).$$

Long inventory lowers both quotes: sell more, buy less.

AS example parameters ($S_0 = 100$, $\sigma = 2$, $\gamma = 0.1$, $k = 1.5$, $A = 140$, $T = 1$, price units), 1,000 paths: mean profit 65.1 with standard deviation 6.5, against 67.4 and 12.7 for the same spread centred on the mid. A little less mean, half the risk.

The exact solution with inventory limits

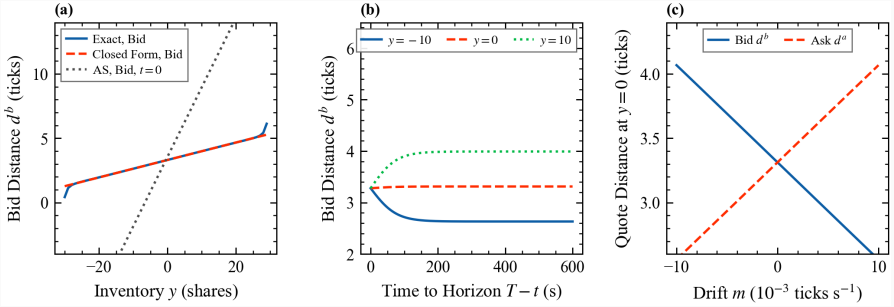
Guéant et al. (2013): the ansatz $u = -\exp(-\gamma(x + yS))\, v_y(t)^{-\gamma/k}$ turns the HJB equation into the **linear** system $\dot{v} = Mv$, $v(T) = \mathbf{1}$, for the vector $v = (v_{-Q}, \ldots, v_Q)$:

$$M_{yy} = \tfrac{k}{2}\gamma\sigma^2 y^2 - k\,m\,y, \qquad M_{y,y\pm1} = -A\big(1 + \tfrac{\gamma}{k}\big)^{-(1+k/\gamma)},$$

$$v(t) = e^{-M(T-t)}\mathbf{1}, \qquad d^b(t,y) = \frac{1}{k}\ln\frac{v_y}{v_{y+1}} + \frac{1}{\gamma}\ln\Big(1+\frac{\gamma}{k}\Big), \qquad d^a(t,y) = \frac{1}{k}\ln\frac{v_y}{v_{y-1}} + \frac{1}{\gamma}\ln\Big(1+\frac{\gamma}{k}\Big).$$

- M is symmetric and tridiagonal: diagonalize once, and quotes at any time cost a few vector operations.
- Long horizons: v aligns with the ground state of M, giving stationary quotes.
- Units: $k\gamma\sigma^2 y^2$ and kmy are in s^{-1}, like A. Every entry of M is a rate.
- Checks: ODE residual below 10^{-6}; terminal quotes; closed form near $y = 0$.

Exact quotes against approximations

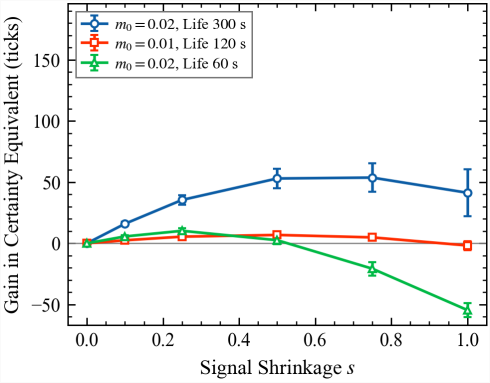


GLFT example: $\sigma = 0.3$ tick s$^{-1/2}$, $A = 0.9$ s^{-1}, $k = 0.3$ tick^{-1}, $\gamma = 0.01$ tick^{-1}, $T = 600$ s, $Q = 30$. (a) Stationary exact quotes match the closed form within 0.001 ticks near $y = 0$; the AS expansion at $t = 0$ overreacts to inventory. (b) Quotes settle after about 200 s. (c) A drift tightens one side.

Strategy	Mean Π_T	Std Π_T	CE
Exact GLFT	1319.2 $\pm$ 1.6	73.6	1291.9
AS approximation	1301.0 $\pm$ 1.5	68.5	1277.7
Symmetric	1324.1 $\pm$ 2.7	120.4	1227.7

2,000 paths; CE: certainty equivalent, $-\frac{1}{\gamma}\ln \mathbb{E}[e^{-\gamma \Pi}]$ for profit Π (ticks).

Quoting with a short-lived signal



The drift is $\pm m_0$ (ticks s^{-1}) and flips sign at rate κ_s (s^{-1}), so the signal's mean life is $1/\kappa_s$. The exact solution treats a drift as permanent. A practitioner plugs in $s\,m(t)$ with shrinkage $s \in [0, 1]$.

- Life 300 s: best $s = 0.75$, gain 53.7 ± 5.8 ticks of CE.
- Life 60 s: raw use **loses** -54.6 ± 2.9; $s = 0.25$ gains 10.2 ± 1.0.

The proper fix is a larger state (signal and inventory) in the HJB equation; shrinkage is a cheap approximation to it. Paired paths, bootstrap errors.

Estimating the fill curve

Data: quote spells i at distance d_i (ticks), each resting for τ_i seconds and ending in a fill ($n_i = 1$) or a cancellation ($n_i = 0$, right-censored). With intensity Ae^{-kd} the log-likelihood is

$$\ell(A, k) = \sum_i \left[n_i(\ln A - k d_i) - A e^{-k d_i} \tau_i \right],$$

a Poisson regression with exposure τ_i. Newton's method on $(\ln A, k)$; standard errors from the Fisher information.

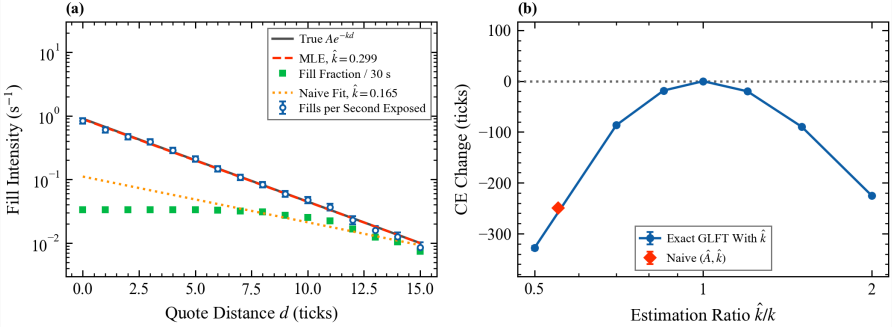
The naive estimate. Divide the fill fraction within a fixed time $t_{\max}$ by $t_{\max}$ and fit an exponential. But a quote fills at most once, so

$$P(\text{fill}) = 1 - e^{-Ae^{-kd} t_{\max}}$$

saturates near the touch. The curve flattens and k is biased down. The censoring that the likelihood handles exactly is ignored.

In practice fills also depend on queue position, which the model leaves out (deck 8).

What a wrong fill curve costs



(a) 6,400 spells, 30 s maximum. MLE: $\hat{A} = 0.886 \pm 0.022$ s^{-1}, $\hat{k} = 0.299 \pm 0.003$ tick^{-1} (true 0.9 and 0.3). Naive fit: $\hat{A} = 0.111$, $\hat{k} = 0.165$.

(b) Exact GLFT quotes computed with a wrong k, run against the true fills (2,000 paired paths). Halving k costs -328.0 ± 1.9 ticks of CE out of 1292.5; doubling it costs -225.1 ± 1.3; the naive $(\hat{A}, \hat{k})$ costs -249.5 ± 1.8.

Evidence, discussion and takeaways

The literature

- AS: inventory-skewed quotes lower profit variance at little cost in mean (Avellaneda and Stoikov, 2008).
- GLFT: exact solution, closed-form asymptotic quotes, drift and inventory limits (Guéant et al., 2013); multi-asset quotes beat independent ones (Guéant, 2017).
- Book-length treatment: Cartea et al. (2015). RL agents tend to beat analytical benchmarks in simulation (Gašperov et al., 2021).

Assumptions to keep in view. Exponential fill curve; fills independent of the next price move (no adverse selection); constant σ, A, k; unit size.

For a liquidity provider

1. Use the exact solution: it is cheap and beats the expansion.
2. Estimate the fill curve by exposure likelihood; naive fill ratios can cost a large share of CE.
3. Shrink signals by their life relative to inventory turnover.
4. The missing piece is adverse selection: fills are not independent of what the price does next (deck 8).

Open question. The optimal map from signal half-life to quotes, in closed form.

Reproduce

Code: `market_making.py`. Experiments: `e7`, `e11`, `e12`, `e21`.

```
from stochastic_lob import market_making as mm
p = mm.MMParams(S0=0, T=600, sigma=0.3, dt=0.2, gamma=0.01, k=0.3, A=0.9)
bid, ask = mm.glft_exact_quotes(Q=30, p=p, taus=[600.0])
tau, filled = mm.simulate_quote_spells(0.9, 0.3, distances, t_max=30.0)
A_hat, k_hat, cov = mm.fit_fill_intensity(distances, tau, filled)
```

References i

Marco Avellaneda and Sasha Stoikov. High-frequency trading in a limit order book. *Quantitative Finance*, 8(3):217–224, 2008.

Álvaro Cartea, Sebastian Jaimungal, and José Penalva. *Algorithmic and High-Frequency Trading*. Cambridge University Press, 2015.

Bruno Gašperov, Stjepan Begušić, Petra Posedel Šimović, and Zvonko Kostanjčar. Reinforcement learning approaches to optimal market making. *Mathematics*, 9(21):2689, 2021.

Olivier Guéant. Optimal market making. *Applied Mathematical Finance*, 24(2):112–154, 2017.

Olivier Guéant, Charles-Albert Lehalle, and Joaquin Fernandez-Tapia. Dealing with the inventory risk: a solution to the market making problem. *Mathematics and Financial Economics*, 7(4):477–507, 2013.